

Genre Classification

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Abstract— This report details four different supervised machine learning models tasked with exploring genre similarity based on audio and playback features. Using a cleaned Kaggle dataset containing 8,520 songs across more than 100 genres, four classification models (K-Nearest Neighbors, Random Forest, Logistic Regression, and Gaussian Naive Bayes) were implemented in scikit-learn. The project specifically focuses on building a binary classifier to determine whether a song belongs to the Hip-Hop genre based on quantitative audio features generated by Spotify's audio analysis models, such as danceability, energy, and valence. This analysis includes detailed procedures for data preprocessing, feature selection, model training, and the evaluation techniques used to compare performance across models. Among all four tested models, the Random Forest classifier achieved the best overall performance; demonstrating the most accurate distinction between Hip-Hop tracks and other genres.

I. INTRODUCTION

Categorizing music by genre is an important task for marketing today for music labels marketing and streaming services. Genre classification traditionally has relied primarily on artists and record labels; this subjective relationship can be influenced by marketing strategies rather than a song's actual sound characteristics. A data-driven approach would provide a more objective way to group and compare songs based on their acoustic properties which could align more with listeners'

music preferences even if the song is originally classified outside their typical listening genre.

The boundary between Hip-Hop and other genres is explored using several supervised machine learning algorithms. While this study focuses on Hip-Hop as the target genre, the same approach could be applied to any musical category. By adjusting the training labels, this can be generalized to explore similarities and overlaps between other genres, offering a scalable method for analyzing relationships across the music landscape.

The four supervised learning algorithms that were implemented consisted of: Random Forest, K-Nearest Neighbors (KNN), Logistic Regression, and Gaussian Naive Bayes (GNB). These models were selected to represent a range of learning approaches. It was expected that non-linear models such as Random Forest and KNN would capture complex interactions between features more effectively, while simpler linear models like Logistic Regression and GNB would provide some variability in the models and could serve as a baseline to compare.

Section II details the cleaning and selection of features for the chosen dataset. Section III outlines the model development process such as hyperparameter tuning, scaling and the built in tools utilized. IV shows the results and compares model performance across accuracy, recall, and AUC metrics, finally, Section V summarizes key

findings, highlights limitations, and proposes methods for extending this work to a broader range of music genres.

II. PREPROCESSING

The original [Kaggle dataset](#) consisted of 10,081 tracks which was slimmed to 8,520 with 1,920 unique artists after removing tracks containing null values. The metrics utilized were:

Danceability-A measure of how suitable a track is for dancing.

Energy-The energy level of the track.

Speechiness-The presence of spoken words in the track.

Acousticness-The acousticness of the track.

Liveness-A measure of the presence of a live audience in the track.

Valence-The musical positiveness or happiness conveyed by the track.

Tempo-The tempo of the track in beats per minute (BPM).

Each metric was computed by Spotify's internal machine learning model which works by analyzing raw audio signals and estimating each of the listed categories.

The features selected for this study consist solely of continuous numerical values representing measurable aspects of each track. Text-based attributes such as track name or artist name were excluded, as they do not contribute directly to the acoustic or rhythmic characteristics that the classifier relies on to evaluate genre similarity.

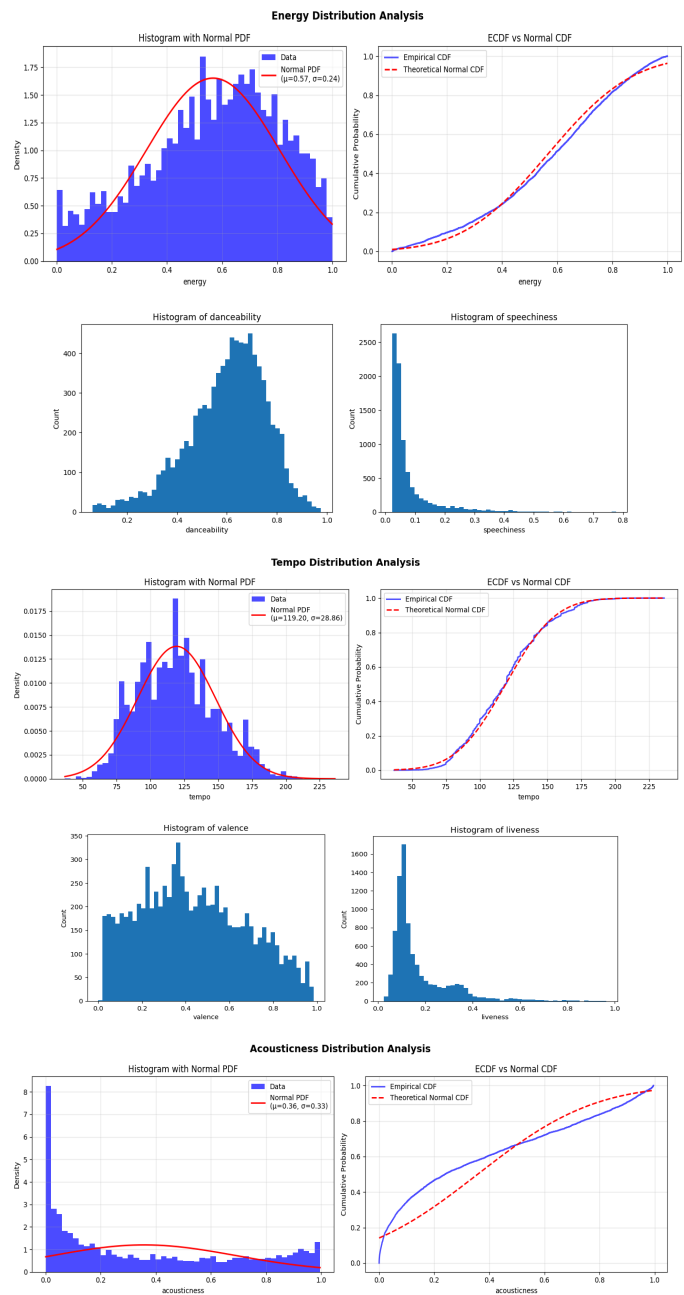


Figure 1: Distribution of Audio Feature Values Across All Tracks With A Normal CDF and Empirical CDF Overlayed

There is a lot of variability within the measured features in Figure 1 with energy, danceability, tempo and valence displaying roughly bell shaped curves indicating moderate similarity to

a normal distribution. In contrast, speechiness, acousticness, and liveness deviate substantially, showing highly skewed or irregular distributions.

To prepare the dataset for binary classification, any track with any mention of “Hip-Hop,” “Hip Hop,” “HipHop,” or “Rap” (case-insensitive) in the genre column was assigned a value of 1, while all other genres were assigned a value of 0. This enabled the model to distinguish Hip-Hop tracks from non-Hip-Hop tracks using a binary target variable.

The dataset also exhibited a significant class imbalance, with Hip-Hop tracks representing approximately 10% of all songs (a 9:1 ratio).

III. METHODOLOGY

To evaluate model performance objectively, the dataset was randomly divided into a training set (80%) and a testing set (20%).

Scaling

Because the numerical features varied in range and magnitude, each feature was standardized using a StandardScaler from scikit-learn to prevent larger values from having a greater influence on the data. This step is particularly important for distance-based algorithms such as K-Nearest Neighbors and for improving stability in models like Logistic Regression.

Hyperparameter Selection

The built in GridSearchCV tool in the Sklearn library was used to finetune the hyperparameters for each model. This allowed a much quicker form of selecting the most optimal option for each parameter.

Optimization was done primarily for recall to address the 9:1 minority ratio of the Hip-Hop class and gathered accuracy, recall, and AUC data on the test set. For tree-based models, `class_weight='balanced'` was applied; for KNN, the hyperparameters `p` and `weights` were tuned; for Logistic Regression, the regularization parameter `C` and solver settings were optimized; and for GNB, the smoothing parameter was adjusted. This procedure minimized the present bias while establishing a consistent evaluation across all models.

Tools

All models were implemented in Python using the scikit-learn 1.7.2 library. Data visualization was performed with pandas 2.3.1, NumPy 2.1, and matplotlib 3.10.7.

Performance Metrics

Model performance was evaluated using several metrics that capture different aspects of classification quality.

Accuracy represents the proportion of songs correctly classified across both categories, providing a general measure of overall performance.

Recall reflects how effectively the model identifies actual Hip-Hop tracks within the dataset.

To further assess ability to discriminate between each genre, the Receiver Operating Characteristic (ROC) curve was used to visualize the models ability at correctly and incorrectly labeling Hip-Hop songs.

The area underneath the curve (AUC) measures how the model separates classes or how confidently it ranks the right songs higher than the wrong ones, across every possible threshold.

Performance was measured on both the training and testing sets to evaluate how well each model generalized to unseen data. A strong performance on the training set with a significant drop on the test set would strongly suggest overfitting, where the model memorizes training examples instead of learning the correct patterns meant to distinguish between classes.

A confusion matrix provides details of a model's predictions, showing the counts of true positives (correctly identified Hip-Hop songs), false positives (non-Hip-Hop songs incorrectly labeled as Hip-Hop), true negatives (correctly identified non-Hip-Hop songs), and false negatives (Hip-Hop songs missed by the model). This visualization helps identify whether errors are biased toward over-prediction or under-prediction of the smaller sized class.

the data and features, which differs from a basic single decision tree algorithm as it helps the model reduce overfitting and capture complex patterns within the dataset. This approach makes the Random Forest effective for classification tasks involving non-linear relationships between features which is perfect for capturing complex characteristics of different music genres.

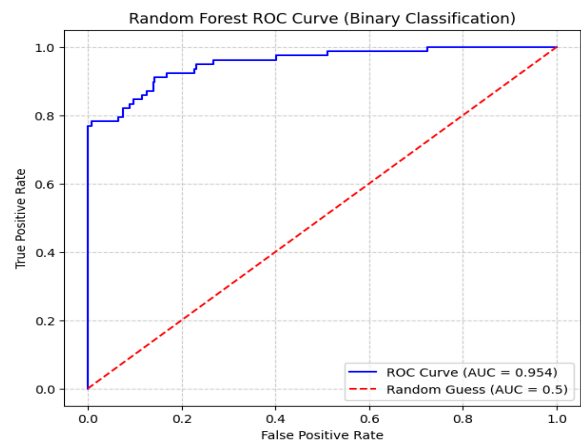


Figure 2. ROC curve for the Random Forest

The Random Forest Model performed extremely well with this dataset with a test accuracy of 97.6% and an AUC of .954. This indicates that it could accurately distinguish whether or not a song was of the Hip-Hop genre or not. The drop in train vs test recall, outlined below in Figure 6, indicates a slight overfitting of the data which means the model does not perform as well on new data but is still highly accurate nonetheless.

IV. PERFORMANCE ANALYSIS

1. Random Forest Classifier

The first model tested was the Random Forest Classifier, which builds multiple decision trees and combines their predictions to improve accuracy and stability. Each tree is trained on random subsets of

2. KNN Classifier

The second model tested was the K-Nearest Neighbors (KNN) algorithm, a distance-based classifier that assigns data points to the majority class of their closest neighbors within the feature space. KNN makes predictions based on similarity, allowing it to adapt well to data with varying patterns and non-linear relationships between

features. This is another highly effective algorithm for genre classifying.

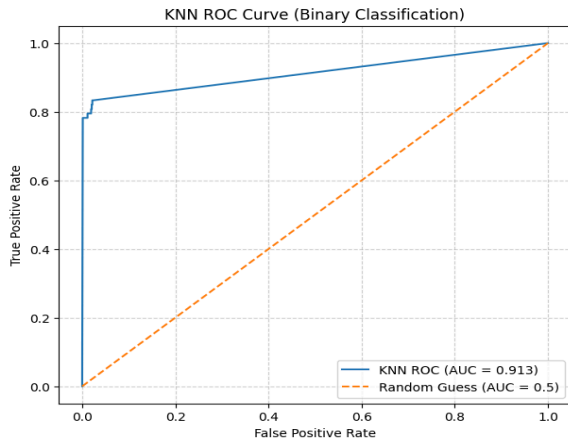


Figure 3: ROC Curve For K-Nearest Neighbors

KNN produced a 96.9% test accuracy and .913 AUC, proving a strong understanding of the relationships between the genres. Its high training accuracy and recall values show that it learns the feature structure very well, although the slight drop in test recall indicates some sensitivity to data imbalance.

3. Logistic Regression

The Logistic Regression model is a linear classifier that estimates probabilities based on weighted combinations of input features. It assumes a linear relationship between features and the odds of it being the target variable. This makes it easier to interpret but is less effective for complex, non-linear feature interactions.

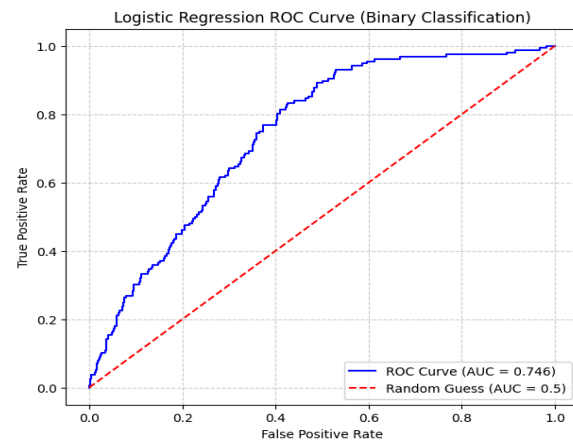


Figure 4: ROC curve for Logistic Regression

The limitations of using the Logistic Regression model meant a lower test accuracy compared to the other methods with a score of 65.4% and an AUC of .756. With corresponding recall values of 0.75 on the training set and 0.71 on the test set, the model performs consistently across both datasets. This small difference between training and testing recall indicates good generalization and suggests that the model is not overfitting. However, the overall performance is moderate compared to more complex models due to the limitations of Logistic Regression mentioned previously thus, the model struggles to fully capture the variation that distinguishes Hip-Hop tracks from other genres

4. Gaussian Naïve Bayes Model

The GBN model assumes that all features within the dataset are independent and normally distributed within each class. While each feature is a numerical continuous value, Speechiness, valence, liveness etc., some values are not completely independent of one another as songs that are high in **energy** likely also have a high **tempo** or **danceability**.

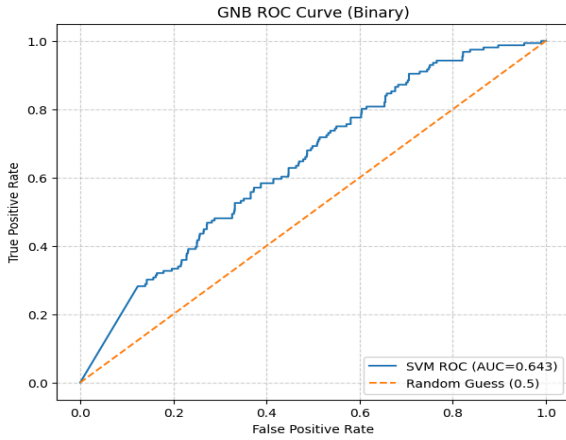


Figure 5: ROC curve for Gaussian Naive Bayes

The GNB model had a moderate performance with a test accuracy of about 75% and an AUC of 0.64. Still producing better results than random guessing but was not as good comparatively at distinguishing between Hip-Hop and non-Hip-Hop tracks. After tuning parameters, the recall rose significantly from 0.019 to .4872 suggesting that while the model could identify some Hip-Hop songs correctly, it often missed subtler examples maybe due to the strong correlations among features that violate the model’s independence assumption.

Figure 6. Training and testing accuracy and recall for all four classification models.

This figure compares the performance of each classifier based on accuracy and recall across both training and testing datasets. It highlights how well each model generalizes to unseen data and illustrates differences in recall, which measures the ability to correctly identify Hip-Hop tracks within an

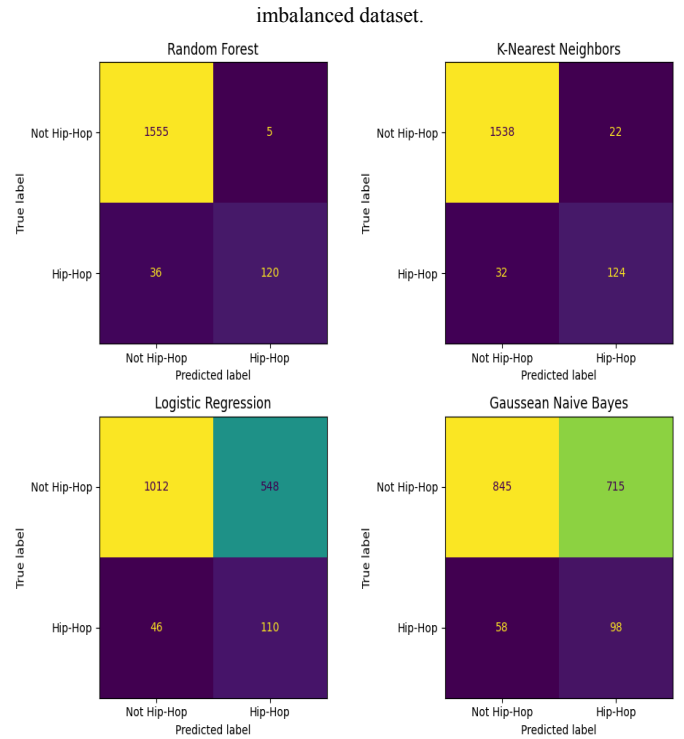


Figure 7: Presents confusion matrices for all four models, revealing the distribution of predictions across true positives, false positives, true negatives, and false negatives.

V. CONCLUSION AND DISCUSSION

This study applied supervised machine learning techniques to classify songs as Hip-Hop or non-Hip-Hop using audio features from the Spotify Song Attributes Kaggle dataset. The goal was to determine whether measurable acoustic characteristics—such as danceability, energy, speechiness, acousticness, liveness, valence, and tempo—could distinguish Hip-Hop tracks from other genres. Four models were trained and tuned using GridSearchCV: Random Forest, K-Nearest Neighbors (KNN), Logistic Regression, and Gaussian Naive Bayes (GNB), with recall prioritized in order to address the dataset’s 9:1 class imbalance.

Among the tested models, Random Forest achieved the highest performance with 97.6% test accuracy and an AUC of 0.954, reflecting its ability to capture complex, non-linear interactions between features. KNN also performed strongly (96.9% accuracy, AUC = 0.913) but showed minor overfitting, while Logistic Regression (65.4%, AUC = 0.756) and GNB (75%, AUC = 0.64) underperformed due to their assumptions of linearity and its limitations of feature independence. Despite high overall accuracy, all models displayed lower recall for Hip-Hop tracks, highlighting the challenge of class imbalance and the tendency to favor the majority class predictions.

These results suggest that the non-linear ensemble and distance-based methods are better suited for capturing the nature of musical features. It could also be true what distinguishes a genre is much more complex than just the audio features and lyrical analysis would provide better insight. Another route to explore this same topic is a multi-class classification network that analyzes the lyrical content, collaboration data, and rhythmic structure. Deep learning approaches trained on raw audio could also be a good attempt at further enhancing performance by learning directly from sound instead of relying on Spotify's API and unpublished machine learning techniques.

In conclusion, this work demonstrates that supervised machine learning can classify genre boundaries using quantitative audio data, with tree-based ensemble methods proving to be the most accurate. Challenges such as class imbalance and feature interdependence were also present; however it symbolises a real world scenario where the selected genre for classification may only make up around 10% of the total music the model will

test on. This framework provides a scalable foundation for understanding stylistic relationships and could be used for supporting music recommendation systems, and exploring the evolving landscape of music.

VI. References

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